Movies, movies, *movies!*

This is an analysis of the "Movies" dataset, provided by the University of Washington's Interactive Data Lab (IDL) and used in several of their Vega-Lite examples.

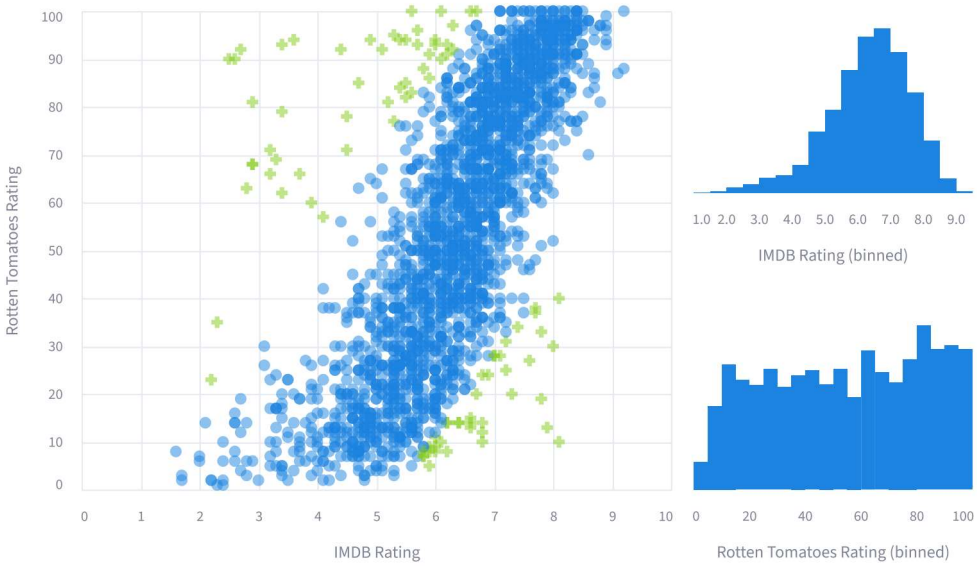
This dataset containers a collection of films and their performance metrics, including box office earnings, budgets, and audience ratings. In order to serve as a teaching resource, this dataset contains known data quality issues. However, it's still a very interesting dataset to play around with, as you'll see below.

Part I: Ratings

How much do critics and viewers agree on ratings? In the analysis below, a movie's Rotten Tomatoes Rating stands for what the professional critics think, while its IMDB Rating stands for what the general public thinks.

Running a linear regression on the data, we find a pretty good correlation between the two variables, though with some prominent outliers (shown with green crosses).

Distribution of ratings from critics vs viewers



Movies most beloved by viewers and disliked by critics

Title	Director	IMDB Rating	Rotten Tomatoes Rating
Aeon Flux	None	8.1	10
The Women	None	7.9	13
The Boondock Saints	None	7.8	19
The Life of David Gale	Alan Parker	7.3	20
Supernova	Francis Ford C	6.8	10
Double Take	None	6.8	12
Hitman	None	6.8	14
Fair Game	None	6.6	13
Crossroads	None	6.6	14
Underworld: Evolution	Len Wiseman	6.6	15

Movies most beloved by critics and disliked by viewers

Title	Director	IMDB Rating	Rotten Tomatoes
Krrish	None	6.1	100
Straight out of Brooklyn	None	5.6	100
Catch Me if You Can	Steven Spielberg	5.7	96
A Nightmare on Elm Street	Wes Craven	5.3	95
The Day the Earth Stood Still	None	5.5	94
House of Wax	None	5.4	94
Superman	Richard Donner	4.9	94
Milk	Gus Van Sant	3.6	94
Beauty and the Beast	Gary Trousdale	3.4	93
Cinderella	None	5.1	92

Part II: Top and bottom-rated directors

Which directors are the most and least beloved? Do critics and viewers agree on this at least?

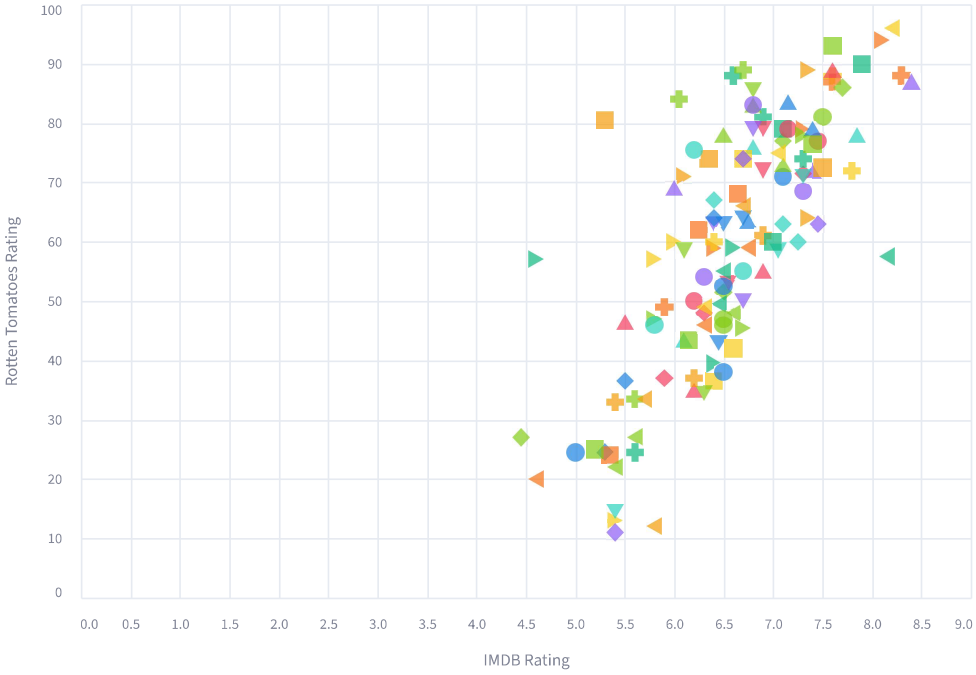
From a quick look at the shape of scatterplot below, even though the median scores for different directions are well-correlated between IMDB and Rotten Tomatoes, the list of top and bottom-rated directors for each are pretty different — especially when considering the exact ordering.

In the visualizations below, **colors** and **shapes** are used to help more quickly identify specific directors across each chart.

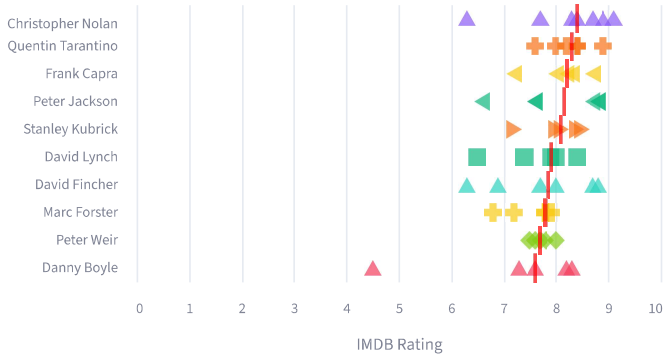
Only consider directors with at least this many movies

5

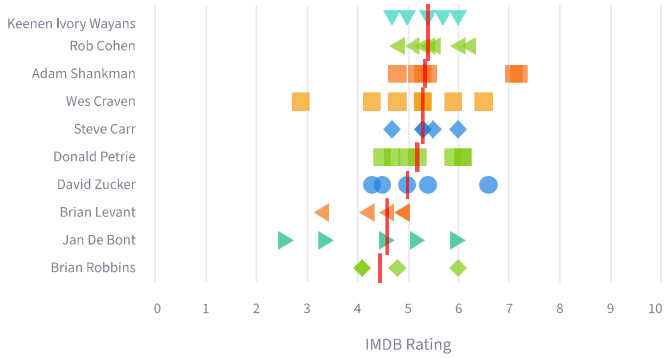
Distribution of ratings from critics vs viewers



Top 10 directors by IMDB Rating

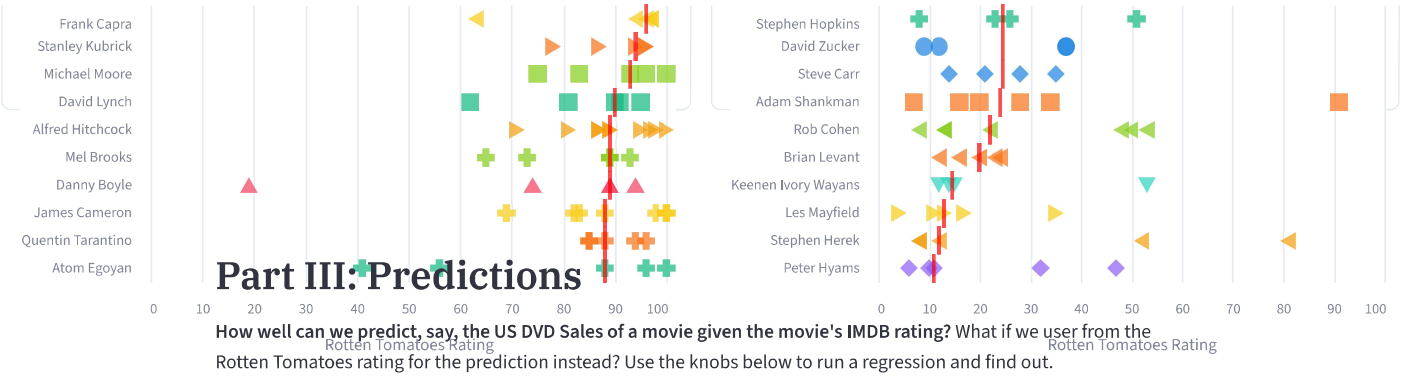


Bottom 10 directors by IMDB Rating



Top 10 directors by Rotten Tomatoes Rating

Bottom 10 directors by Rotten Tomatoes Rating



Part III: Predictions

How well can we predict, say, the US DVD Sales of a movie given the movie's IMDB rating? What if we user from the Rotten Tomatoes rating for the prediction instead? Use the knobs below to run a regression and find out.

The blue line is the model prediction, and outliers are shown as green crosses.

X Axis (predictor)

IMDB Rating

Y Axis (target)

US DVD Sales

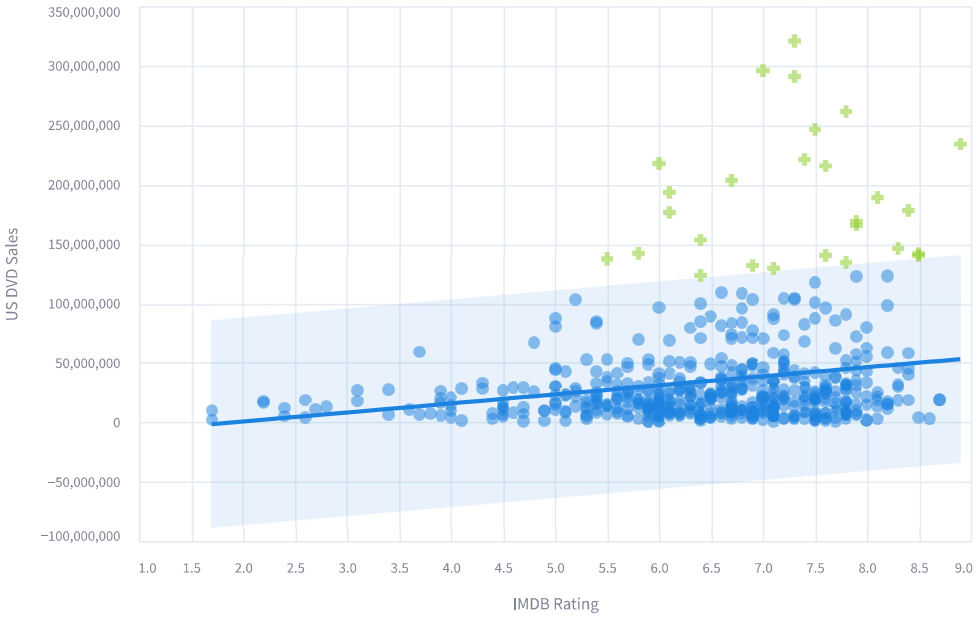
Confidence interval (sigma)

2.00

Regression type

Linear regression | LOESS regression

US DVD Sales by IMDB Rating



Outliers

These are the movies that do not fit the model.

Number of outliers

27

Title	Director	IMDB Rating	Rotten Tomatoes Rating
300	Zack Snyder	7.8	60
Alvin and the Chipmunks	Tim Hill	5.5	27
Avatar	James Cameron	8.3	83
The Dark Knight	Christopher Nolan	8.9	93
Cars	John Lasseter	7.5	74
Chicken Little	Mark Dindal	5.8	36
The Departed	Martin Scorsese	8.5	93
The Hangover	Todd Phillips	7.9	78
Harry Potter and the Goble	Mike Newell	7.6	88
Harry Potter and the Order	David Yates	7.4	78
Happy Feet	George Miller	6.7	74
Ice Age: The Meltdown	Carlos Saldanha	6.9	
Iron Man	Jon Favreau	7.9	94
King Kong	Peter Jackson	7.6	83
I am Legend	Francis Lawrence	7.1	69

Part IV: Browse the full dataset

Title	US Gross	Worldwide Gross	US DVD Sales	Production Budget	Release Date	MPAA Rating	Running Time min	Distributor	Source	Major G
The Land Girls	$146,083.00	$146,083.00	None	$8,000,000.00	Jun 12 1998	R	None	Gramercy	None	None
First Love, Last Rites	$10,876.00	$10,876.00	None	$300,000.00	Aug 07 1998	R	None	Strand	None	Drama
I Married a Strange Person	$203,134.00	$203,134.00	None	$250,000.00	Aug 28 1998	None	None	Lionsgate	None	Comed
Let's Talk About Sex	$373,615.00	$373,615.00	None	$300,000.00	Sep 11 1998	None	None	Fine Line	None	Comed
Slam	$1,009,819.00	$1,087,521.00	None	$1,000,000.00	Oct 09 1998	R	None	Trimark	Original Sc	Drama
Mississippi Mermaid	$24,551.00	$2,624,551.00	None	$1,600,000.00	Jan 15 1999	None	None	MGM	None	None
Following	$44,705.00	$44,705.00	None	$6,000.00	Apr 04 1999	R	None	Zeitgeist	None	None
Foolish	$6,026,908.00	$6,026,908.00	None	$1,600,000.00	Apr 09 1999	R	None	Artisan	Original Sc	Comed
Pirates	$1,641,825.00	$6,341,825.00	None	$40,000,000.00	Jul 01 1986	R	None	None	None	None
Duel in the Sun	$20,400,000.00	$20,400,000.00	None	$6,000,000.00	Dec 31 2046	None	None	None	None	None
Tom Jones	$37,600,000.00	$37,600,000.00	None	$1,000,000.00	Oct 07 1963	None	None	None	None	None
Oliver!	$37,402,877.00	$37,402,877.00	None	$10,000,000.00	Dec 11 1968	None	None	Sony Pictures	None	Musica
To Kill A Mockingbird	$13,129,846.00	$13,129,846.00	None	$2,000,000.00	Dec 25 1962	None	None	Universal	None	None
Tora, Tora, Tora	$29,548,291.00	$29,548,291.00	None	$25,000,000.00	Sep 23 1970	None	None	None	None	None
Hollywood Shuffle	$5,228,617.00	$5,228,617.00	None	$100,000.00	Mar 01 1987	None	None	None	None	None